

Ideas Imported from the STFT Nicola–Tilli Stability Paper

This note records what can be imported from `s00222-024-01248-2-3.pdf`, the 2024 Inventiones paper

Gómez–Guerra–Ramos–Tilli, *Stability of the Faber–Krahn inequality for the short-time Fourier transform*.

The paper proves a sharp quantitative stability version of the Nicola–Tilli Faber–Krahn theorem for the short-time Fourier transform. It is not directly a torsion result, but several mechanisms are relevant to `plan2.md`.

1 The Structural Dictionary

In the STFT/Fock setting, the central object is

$$u_F(z) = |F(z)|^2 e^{-\pi|z|^2}$$

and its superlevel sets

$$A_t = \{u_F > t\}.$$

The rearranged profile $u^*(s)$ satisfies a one-dimensional convexity inequality. The optimizer profile is e^{-s} . Stability is obtained by quantifying the area between $u^*(s)$ and e^{-s} , and then pushing the information back to sets.

For torsion, the analogous objects are

$$u = u_\Omega, \quad E_t = \{u > t\}, \quad v(s) = u_\Omega^*(s),$$

and the optimizer profile is the ball profile

$$b(s) = u_B^*(s) = \frac{R^2 - (s/\omega_n)^{2/n}}{2n}, \quad |B| = \omega_n R^n.$$

The torsion level-set inequality gives

$$v'(s) \geq b'(s).$$

Thus

$$h(s) := b(s) - v(s)$$

is nonnegative and nonincreasing, and

$$\int_0^{|\Omega|} h(s) ds = \int_B u_B - \int_\Omega u_\Omega = 2(E(\Omega) - E(B)).$$

This is the exact torsion analogue of the STFT paper’s “area between profiles” method.

2 Immediate Lemma: Profile Deficit Controls Level Heights

Since h is nonincreasing,

$$0 \leq h(s) \leq \frac{1}{s} \int_0^s h(r) dr \leq \frac{2(E(\Omega) - E(B))}{s}.$$

Therefore, for $s \in [\theta|\Omega|, |\Omega|]$,

$$|u_B^*(s) - u_\Omega^*(s)| \leq \frac{2(E(\Omega) - E(B))}{\theta|\Omega|}.$$

This gives quantitative control of near-boundary level heights. If $s = L - \eta$ and $E(\Omega) - E(B) \ll \eta$, then the level

$$t_\eta := u_\Omega^*(L - \eta)$$

is positive and comparable to the corresponding ball level. This selects a boundary-layer set

$$E_{t_\eta} = \{u > t_\eta\}$$

with exactly

$$|\Omega \setminus E_{t_\eta}| = \eta.$$

This should be combined with the weighted level-set deficit identity to find near-boundary levels that are also geometrically good.

3 Superlevel Replacement

The STFT proof does not try to control an arbitrary set directly. It first replaces the set Λ by the matched superlevel set

$$A_\Lambda = \{u_F > u_F^*(|\Lambda|)\}.$$

Then it proves:

1. the deficit controls the function's distance to the optimizer;
2. this makes A_Λ regular and close to a ball;
3. a transport argument controls $|\Lambda \Delta A_\Lambda|$;
4. hence Λ itself is close to a ball.

For torsion, the analogous move is:

1. choose $E_t = \{u_\Omega > t\}$ with prescribed volume s ;
2. prove E_t is close to a ball using the level-set identity;
3. use

$$\mathcal{A}(\Omega) \leq C \frac{|\Omega \setminus E_t|}{|\Omega|} + C\mathcal{A}(E_t)$$

to transfer the result back to Ω .

The point of the profile-gap lemma is that it chooses near-boundary levels with controlled height, while the exact level-set identity supplies averaged control of $D_H(t) + D_I(t)$.

4 Transport Idea for the Set Comparison

In the STFT paper, set stability is proved by transporting mass from $A_\Lambda \setminus \Lambda$ to $\Lambda \setminus A_\Lambda$. Points moved farther away from the center lose a definite amount of Gaussian weight, so the concentration deficit controls the amount of badly moved mass.

For torsion, a literal radial transport is unavailable for arbitrary domains. But for selected minimizers or near-ball sets, a normal-transport analogue should be possible:

- compare a near-boundary level E_t with Ω along the flow lines of ∇u , or along the free-boundary normal when Ω is already a selected minimizer;
- use the nondegeneracy

$$c \operatorname{dist}(x, \partial\Omega) \leq u(x) \leq C \operatorname{dist}(x, \partial\Omega)$$

near the free boundary to convert height t into boundary-layer thickness;

- use density/perimeter bounds to estimate the symmetric difference.

This is one reason the hybrid route looks more promising than the pure level-set route: the selected minimizers have exactly the free-boundary structure needed to make this transport quantitative.

5 Geometry of Superlevel Sets

The STFT proof obtains star-shapedness, smoothness, and convexity of relevant superlevel sets from closeness of F to the Gaussian and analyticity.

For torsion, the replacement should be:

- for arbitrary domains, no comparable regularity is available;
- for selected minimizers, Alt–Caffarelli regularity gives $C^{1,\gamma}$ free boundary once flatness is known;
- after entering the graph regime, Schauder estimates give u_Ω close to u_B , and then regular near-boundary level sets follow from the implicit function theorem.

So the STFT “regular superlevel geometry” mechanism should be imported only after the BDV selection step.

6 Variational/Fuglede-Style Route

The STFT paper also proves stability by a variational second-variation argument for

$$K[F] = \frac{I_F(s)}{\|F\|^2},$$

where $I_F(s)$ is the integral of u_F over its superlevel set of measure s . Around the optimizer $F \equiv 1$, after removing neutral modes, the second variation is uniformly negative.

The torsion analogue would be to study, near the ball,

$$\mathcal{K}_s(\Omega) := \int_{\{u_\Omega > u_\Omega^*(s)\}} u_\Omega \, dx$$

under nearly spherical perturbations of Ω . The expected result is a second-variation estimate of the form

$$\mathcal{K}_s(B) - \mathcal{K}_s(\Omega) \gtrsim_s \|\varphi\|_{H^{1/2}(\partial B)}^2$$

after imposing volume and barycenter orthogonality. This would be a level-set-sensitive analogue of BDV's nearly spherical expansion for E .

This is probably harder than the profile-gap lemma, but it could produce a direct near-ball stability theorem for the level-set route.

7 Revised Route for plan2.md

The most promising revised route is:

1. Prove the finite-perimeter level-set identity.
2. Add the profile-gap lemma $0 \leq b(s) - v(s) \leq 2\delta_E/s$.
3. Choose boundary-layer volumes $s = L - \eta$, with $\delta_E \ll \eta \ll 1$.
4. Use the weighted deficit identity to find one such level with small $D_H + D_I$.
5. Apply quantitative isoperimetry to D_I , and the weighted variance identity to D_H .
6. For arbitrary domains this still leaves a regularity gap; for selected minimizers, use the BDV free-boundary estimates to close it.
7. Transfer from E_t back to Ω by the boundary-layer asymmetry lemma.

This is a genuine import from the STFT stability paper: first stabilize the matched superlevel set, then transfer stability back to the original set.

8 Analytic/Native-Space Refinement

The GGRT proof benefits from living in the Bargmann–Fock space. For torsion, there is no global entire structure for arbitrary domains, but there is a native harmonic substitute:

$$h_\Omega(x) := u_\Omega(x) + \frac{|x|^2}{2N}, \quad \Delta h_\Omega = 0.$$

For a ball, h_B is constant. For a nearly spherical domain, after pulling back to the ball, the first variation of $h_\Omega - h_B$ is the harmonic extension of the boundary graph. This is the natural space in which to try a GGRT-style second variation.

In dimension two there is an even more literal analytic object:

$$F_\Omega(z) := \partial_z u_\Omega(z) + \frac{\bar{z}}{4}.$$

Since $\partial_{\bar{z}}\partial_z u = -1/4$, the function F_Ω is holomorphic in Ω , and $F_B \equiv 0$ for disks centered at the origin. For selected smooth near-disks, pulling F_Ω back to the unit disk could put the problem into a Bergman/Hardy-type setting, closer in spirit to the Fock-space proof.

The concrete calculation to try is the second variation at the ball of

$$\mathcal{K}_s(\Omega) = \int_{\{u_\Omega > u_\Omega^*(s)\}} u_\Omega \, dx.$$

If the resulting quadratic form is diagonal in spherical harmonics and has a spectral gap after removing the $k = 0$ and $k = 1$ modes, this would give a native torsion analogue of the GGRT variational proof.